

consider when selecting an ESD-control floor. The first and foremost is whether to opt for a conductive or static dissipative system.

A conductive material, because of its low electrical resistance, allows electrons to flow easily through it or across its surface. When a conductive material becomes charged, the charge gets uniformly distributed across the surface of the material.

If the charged conductive material makes contact with another conductive material, electrons move between the materials easily. If the second conductor is attached to an AC equipment's ground or any other grounding point, electrons flow to ground and excess charge on the conductor gets neutralized.

Static dissipative materials have electrical resistance between insulative and conductive materials. There can be electron flow across or through the dissipative material, but surface resistance or volume resistance of the material controls it.

Charge can be generated triboelectrically on a static dissipative material. However, like the conductive material, a static dissipative surface allows the transfer of charge to ground or other conductive objects. Transfer of charge from a static dissipative material generally takes longer than from a conductive material of equivalent size. Charge transfers from static dissipative materials are significantly faster than from insulators, and slower than from conductive substances.

Which control technique to use out of the two largely depends on the application. In a general office environment, a coating with antistatic characteristics that reduces triboelectric charging on contact surfaces will normally be sufficient.

However, in an area where any significant charge accumulation may damage sensitive components or create a spark risk, emphasis is on dissipating electrostatic charge before it can build up to dangerous levels.

For example, conductive flooring (25,000-ohm to 100,000-ohm resistance) is commonly used in facilities such as ammunition manufacturing, while electronics manufacturing will most often specify flooring in the dissipative range of 10^6 to 10^9 ohms.

The amount of resistance in a static-control floor determines the amount of electrical current that can pass through the floor. As resistance decreases, electrical current increases. In practice, ESD control is usually achieved by a combination of static dissipative footwear and flooring, so that there is an adequately low-resistance path from the person via footwear and flooring to ground (zero/earth potential). In many cases, dissipative flooring is the most cost-effective option.

Flooring standards

The types of antistatic flooring are categorized into several standards by ESD/EOS Association Inc. The following points identify the main categories and key criteria that a floor has to fulfil in order to meet these:

1. ANSI/ESD S20.20 (S20.20) requires a system resistance of a person through the floor and to ground of less than 3.5×10^9 ohms (35 megaohms) when the floor is to be used as the primary ground. Many dissipative floors have resistance ranges above 35 megaohms before considering operator resistance.

2. ANSI/ESD S20.20 revised in 2014 for ESD workplace compliance mentions three standards within the specification relating to flooring materials; all three standards are required to be met for compliance.

3. ANSI/ESD STM 7.1 Floor Materials 'Resistive Characterisation of Materials' states that any flooring material must measure below 1.0×10^9 ohms to ground.

4. ANSI/ESD STM 97.1 Floor Materials and Footwear 'Resistance in Combination with a Person' states that the recommended maximum system resistance is 3.5×10^9 .

5. ANSI/ESD STM 97.2 Floor Materials and Footwear 'Voltage Measurement in Combination with a Person' suggests that the recommended maximum voltage allowed is 100 volts.

Selection of an effective ESD floor-covering system should always begin with a thorough evaluation of the intended use and possible future use. Intended use should include evaluation of devices or processes in the environment that are most sensitive to ESD events, chemical resistance and aesthetic requirements.

Understanding the need for and importance of ESD-compliant footwear is also critical in the performance of the ESD flooring chosen. Grounding personnel require effective ESD footwear that is appropriate to the application, properly worn and in good repair.

A variety of antistatic flooring systems are available for static-sensitive environments that allow architects and consultants to meet the specific needs of the work area in question. This comprehensive product range includes floor finishes for production and assembly rooms, storage areas, office space, R&D facilities or clean rooms, to name a few.

These surfaces do not require periodic treatment to maintain antistatic control capability that complies with the latest ANSI/ESD standards.

Ensuring that the floor area is tailored to the level of electrostatic charge that is likely to be generated within an environment with sensitive electronic equipment is key to avoiding failures and resultant business issues that can stem from the charges sparking into devices and ruining components within.

When specifying an antistatic floor finish, it is important to work with both a supplier and applicator who are well versed with the facts around electrostatic charge buildup in the floor and how this risk can be effectively reduced. **EFY**